Posa Harsha Vardhan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.64
Sri Chaitanya College	Class XII (Andhra Pradesh Board of Intermidiate Education)	2022	98.00%
Sri Chaitanya school	Class X (Board of Secondary Education Andhra pradesh)	2020	99.67%

EXTRACURRICULAR/CLUB PORS

NSS Executive Leadership (Volunteer -> Student Representative -> General Secretary -> Mentor)

- Organized over 40 Swachh Bharat clean drives, 15 civic awareness campaigns, 5 orphanage visits, 10+ plantation drives and 4 blood donation camps across the campus & nearby villages leading and co-ordinating 1000+ NSS volunteers.
- Partnered with 8 regional government schools around IITH to conduct Vidyaadaan teaching sessions with interactive methods and 30+ motivational campus exposure tours for Government school students.
- **Founded NSS's first residential rural-development camp**, leading a 9-member team for a 3 day residential camp in Jukal village to identify education, healthcare and agricultural challenges and synthesize findings for submission to the Sangareddy District Collectorate (Ms. Uma Harathi IAS).
- **Conceptualized and executed Re-Ignite**, NSS's first 7-day flagship event, featuring national social-impact leaders(Padma Awardees) and campus-wide initiatives including CPR/First Aid training for **450+ participants** and the first Indian Sign Language workshop at IITH.

NGO Collaborations & Social Impact Drives (NSS)

- Coordinated a lake remediation and cleaning drive at Barla Kunta Lake in Hyderabad with **'Connect For'** NGO and discussed major pollution causes in March 2024
- Partnered with **'Vriddhi Care'** to visit old age homes and distribute essential sanitary pads and daily necessities to senior citizens.
- Assisted **'Research Heights'** NGO for the SDG Walk 2047 along Necklace Road, involving 30+ educational institutions across Hyderabad.

Interests & Personal Pursuits

Puzzle Solving: Can Solve 7 variants of Rubik's Cube; enjoy algorithmic and spatial problem solving.

Politics and Mythology: Strong interest in mythology and politics and most likely to start an argument with friends.

Personal Skills: 3 ball juggler, Beginner-level flute learner and recreational skater.

SCHOLASTIC ACHIEVEMENTS

First Prize in Ansys + Infinitipoint Simulation Hackathon

- Won the first prize along with a voucher worth of INR 10,000 at the simulation competition.
- Organized by Ansys and Infinitipoint at IIT Hyderabad focused on photonic circuit design & simulation using Ansys Lumerical software
- Successfully won the competition as an undergraduate student, competing directly against senior PG and Ph.D. scholars

Journal Paper Submission in Springer Nature under Optical and Quantum electronics

- Co-author of a research manuscript titled "Optimization of photonic waveguide bends for low-index contrast material platforms," submitted to Springer Nature's Optical and Quantum Electronics journal

- Research focuses on reducing waveguide footprints on the InP platform, helping optimize ring resonators for narrow line width lasers for Photonic Integrated Circuits (PICs).

POSITIONS OF RESPONSIBILITY

Projects core, Prakrithi

Class Representative, Department of Electrical (IC Design and Technology)

EXPERIENCE

Teaching Assistant - Basic Circuits (NPTEL)

- Prepared assignments and question papers to test and improve student understanding of basic circuit concepts for the students.
- Helped students regularly by clearing their doubts and giving clear explanations during the course

PROJECTS

Hands-on Training on Semiconductor Fabrication | Ministry of Education w/ IITH

- 10 day hands-on training in Purdue University Bircks Nano Centre.
- 21 day hands-on use of manufacturing tools for semiconductor fabrication on the IIT Hyderabad campus.
- Learnt cutting-edge technology in the fabrication of FinFETs, CFETs, NWFETs and MOSFETs

Academic Projects

- ADC & DAC Designed and implemented a 3-bit Flash Analog-to-Digital Converter (ADC) and 3-bit R-2R Ladder based Digital to Analog Converter (DAC) using discrete components and breadboards.
- Digital Clock Developed a digital clock display using seven-segment LED displays, discrete logic gates, and breadboard
- Designed and Implemented SRAM cell on breadboard
- Designed Verilog modules and Testbenches using iverilog

Analog Circuit Design

- Designed single stage amplifiers, current mirrors, band-gap references and 5-transistor OTA in Cadence Virtuoso.

Simulations and Analysis of MOSFETs and Quantum well structures

- performed simulations on long and short channel mosfets using in house TCAD tool and analyzed the electron concentration and potential across the mosfet
- performed simulations on a DGFET which solves 1D-Schrodinger equation and analyzed how gate tunneling, eigen energies and peak carrier concentration will get affected by the semiconductor thickness and effective mass
- Analyzed the mosfet characteristics down into the cryogenic temperature ranges